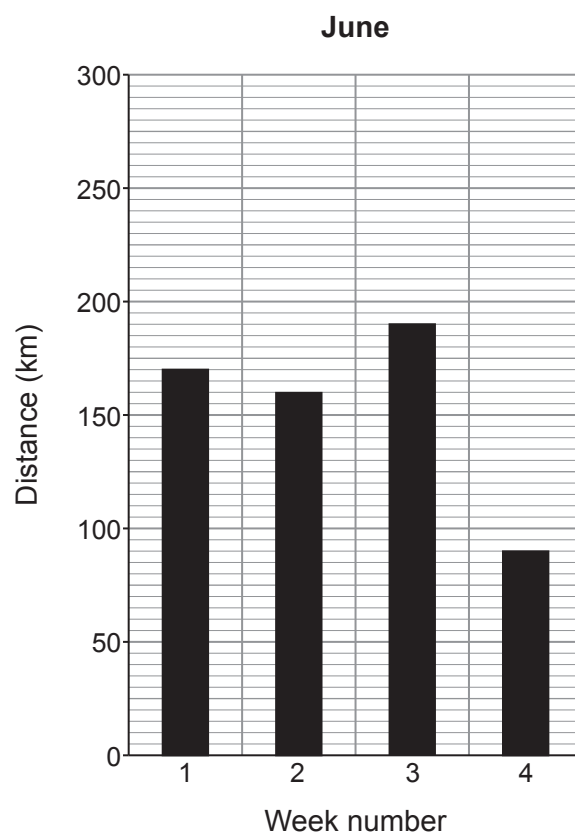
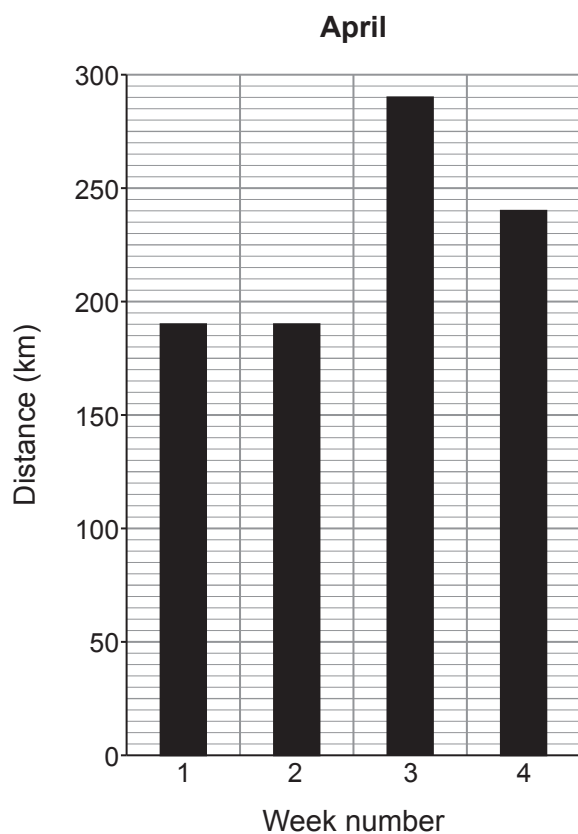


4. A fitness app is used by a cyclist to monitor his cycling performance.

- (a) The distance cycled each week is displayed as a chart.
The data for two months, April and June, are shown below.



- (i) The total distance travelled by the cyclist in June was 610 km.
Show that the total distance travelled by the cyclist in April was 910 km.

[1]



- (ii) The cyclist cycled for a total time of 31.5 hours during April and 19.25 hours during June.
A student suggests that the mean speed of the cyclist is greater in June than in April.
Use an equation from page 2 to determine whether the student is correct. [3]
Space for calculations.

.....
.....

- (b) (i) A track cyclist has an initial velocity of 8.0 m/s.
She uniformly accelerates towards the finish line, travelling a distance of 42 m, in a time of 3.0 s. [4]

Use the equation:

$$x = ut + \frac{1}{2}at^2$$

to calculate the cyclist's acceleration and state its unit.

acceleration =

unit =

- (ii) The resultant force that causes the acceleration towards the finish line is 284 N.

- I. Use an equation from page 2 to calculate the mass of the bike and the cyclist. [2]

mass of bike and cyclist = kg



II. The mass of the cyclist is 64.5 kg. Calculate the mass of the bike.

[1]

Examiner
only

mass of bike = kg

11

